

Student Assessment Information Sheet

This document is for University of Dundee students for their use in completing assessed work. It should not be passed to third parties or posted on any other website.

Assessment Information	
Module Name:	DevOps and MicroServices
Module Code:	AC51041/GA42010
Assessment Title:	System Implementation and Continuous Deployment
Assessment Weighting:	80%
Assessment Due Date	Sunday, 21 st December 2025

If you have questions about this assessment, please contact one of the staff on the module

Learning Outcomes
This assessment addresses the following learning outcomes of the module: - Compare and contrast technologies used to develop microservices systems: You will document the technologies (such as containerisation, cloud services, databases, etc.) used in your implementation and compare alternatives where appropriate. Explain why your choices are the most suitable for the system. - Understand the principles of continual development: You will demonstrate the continual integration and deployment process in your video, showing how code changes flow through the CI/CD pipeline, from development to deployment, with emphasis on automation and continuous delivery. - Develop a system using NoSQL: You will integrate a NoSQL database into your microservices system and demonstrate its functionality. The video should clearly show data operations (such as adding or retrieving data) from the NoSQL system as part of the demo. - Implement a variety of tools to develop a cloud-based microservices system: You will deploy a microservices system using a variety of cloud and containerisation tools. The report should cover the tools used (e.g., Docker, AWS, CI/CD, etc.), while the video should demonstrate the system working, and show how the services interact. - Develop and apply problem-solving, team working, communication, time-management, self-assessment, and independent study skills: You will showcase your ability to solve complex problems encountered during development, managing your time effectively. This will be assessed through the clarity of your report and video, as well as your reflection on challenges and solutions during the development process.

Given the broad and evolving nature of these tools, it is important to be clear about what is permitted in each assessment. Your module lecturers will indicate to what extent generative AI can be used. If you are unsure whether the use of a specific AI tool is permitted, please contact your module coordinator for guidance. Misuse or failure to disclose AI assistance may be treated as academic misconduct and subject to procedures outlined in the [academic misconduct by students code of practice](#).

The levels of AI usage in assessment are:

RAG Rating	Permitted Level	Description
Red (Forbidden)	No	AI tools must not be used for this assessment. This includes generating content, conducting analysis, or providing answers to specific questions.
Amber (Restricted)	No	AI tools may be used with caution, but their use must be explicitly acknowledged. For example, AI assistance in brainstorming, drafting, or structuring ideas is allowed, but the final submission must be your own work. Any AI-generated content must be fully reviewed, edited, and critically evaluated.
Green (Permitted)	Yes	AI tools are permitted for this assessment without restriction, but their use must be explicitly acknowledged. You may use AI systems for research assistance, content generation, and other purposes, provided that the usage aligns with the learning outcomes and your understanding is demonstrated. You should engage critically with output and apply your own judgement.

Assessment Task

In assessment 1, you were tasked to design a scalable conference room booking system by a company that provides conference rooms services in various locations across the country (see assessment 1 for details about that scenario).

In this assessment, you are required to implement that system. You will develop, deploy, and showcase the system using cloud-based microservices principles, as well as Continuous Integration and Continuous Deployment (CI/CD) practices.

Your task is to:

1. Implement the System:

- Use a cloud infrastructure (AWS recommended) and containerisation to implement the system using a microservices approach (see assessment 1 for details).
- Ensure that all services (authentication, weather forecast, etc.) are properly

implemented and integrated with relevant cloud services.

- Implement the NoSQL database for managing relevant data and/or other services as discussed in your design.

2. Set Up a CI/CD Pipeline:

- Create a CI/CD pipeline that will automate the testing, integration, and deployment of your microservices.
- Ensure that your CI/CD pipeline is well-documented and functional, with automated testing to validate microservice functionality before each deployment.

3. Showcase Your System in a Video Demo (5 - 8 minutes long) where you:

- Explain the system architecture and cloud services used.
- Demonstrate the functionality of the system, showing all key features of the system.
- Show the **CI/CD pipeline in action** (e.g., by making a code change, testing it, and pushing it through the pipeline to show deployment).
- Briefly discuss any challenges or trade-offs you encountered during implementation.
- **Please note:** If your video is longer than 8 minutes then only the first 8 minutes will be watched and graded.

4. Submit an Implementation Report (maximum of 6 pages) that includes:

- A description of the system's architecture, focusing on what was implemented and how it was built.
- An explanation of the CI/CD pipeline: the tools used, the process of setting up, and the automation implemented.
- Screenshots and/or logs showing your system running and proof of a successful deployment via the CI/CD pipeline.
- Reflections on the challenges faced during implementation, and how these were resolved.

Deliverables

Video demo (40%):

A 5-8 minute video demonstrating:

- The **running system** showing all key features.
- A **live demonstration of the CI/CD pipeline** (e.g., pushing a code change through the pipeline).
- A brief explanation of the **architecture** and **technologies** used.
- **Challenges encountered** during the development process.
- **CI/CD Pipeline** plan with tools, processes, and rationale.

Implementation Report (60%):

A maximum 6-page report (excluding references and appendices) that includes:

- A description of the **architecture** and **technologies** implemented.
- A **detailed explanation** of the **CI/CD pipeline**, including the steps taken to set it up and how it functions.
- **Screenshots** of the system and logs showing deployment success.
- **Reflections on challenges and solutions** during development.
- A link to your video should be included at the top of your report. You should ensure that the link works properly.

The report must be submitted as a word or PDF document.

How to Submit

Assessments should be submitted to My Dundee in the 'Assessment and Feedback' area. Please only submit a single word or PDF document.

Assessment Feedback

Feedback on your assessment will be provided via **My Dundee** and will be available within **three weeks** of the submission deadline. Any alteration to this will be communicated with you in advance through My Dundee.

Feedback can be provided in multiple ways. This can include informally during lab and tutorial sessions (e.g. discussing work and sample solutions being explored), and formally after marking (e.g. written, oral, video-based).

Academic Integrity

By submitting this Assessment, you are bound by the University of Dundee policies in relation to Academic Standards and Integrity. Please refer to [Policy Statements, Regulations and Guidance](#) for more information

Late Penalties

One grade point per day late (meaning if a submission is one day late and marked as a C2 it will receive a C3 grade). A day is defined as each 24-hour period following the submission deadline including weekends and holidays. Assessments submitted more than 5 days after the agreed deadline will receive a zero mark (AB).

Generic Assessment Criteria		
A1 – A5	Meets learning outcomes	Excellent work demonstrating a comprehensive understanding, creativity, and advanced critical/analytical abilities. Substantial research well beyond the recommended minimum, using diverse methodologies. Exceptional grasp of knowledge and discipline-specific theories. Shows creative thinking, originality, and independent work. Exemplary use of learning resources. Exhibits highly developed problem-solving skills. Work produced with exceptional accuracy, proficiency, and independence. Superior communication and expression.
B1 – B3		Very good work, showing a solid understanding of theories, concepts, and issues, with clear critical analysis. Research goes beyond the recommended minimum, using established methodologies accurately, with minimal irrelevant content. Very good understanding of knowledge and discipline-specific theories, with some originality and independence. Very good use of learning resources. Displays well-structured arguments and interpretation. Demonstrates creativity, originality, and problem-solving abilities. Work produced with high accuracy, proficiency, and independence. Effective communication and expression, with a solid professional skill set.
C1 – C3		Good work showing an understanding of relevant theories, concepts, and issues, with some critical analysis. Research is carried out accurately using established methodologies, occasionally extending beyond the minimum required. Some errors or irrelevant content may be present. A good understanding of knowledge and discipline-specific theories, with some signs of originality and independence. Good use of learning resources. Provides logical arguments and interpretation with supporting evidence. Shows some creativity and problem-solving skills, though inconsistencies may appear. Expression and presentation are generally accurate, proficient, and carried out with some independence. Good communication and expression with an appropriate professional skill set.
D1 – D3		Adequate work showing a basic understanding of relevant theories, concepts, and issues. Demonstrates sufficient understanding of knowledge and discipline-specific theories to address key concepts. Some errors or gaps in research may be present, and irrelevant content is likely. Basic use of learning resources. Shows some ability to formulate and sustain an argument. Displays some creativity and problem-solving abilities, though inconsistencies are present. Expression and presentation meet minimum standards for accuracy. Communication and expression are sufficient.
MF1 – MF3	Fails to meet learning outcomes	Does not meet the required outcomes. Shows a limited understanding of relevant theories, concepts, and issues. Minimal evidence of research or use of established methodologies. Some relevant content may be included, but there are significant gaps in analysis. Fundamental errors and misunderstandings are likely. Limited use of learning resources. Arguments are weak and poorly developed. Little originality, creativity, or problem-solving ability is evident. Expression and presentation fall short of accuracy and proficiency. Communication and expression are insufficient, with gaps in professional skills.

BF - CF		Does not meet outcomes. Demonstrates a lack of understanding of relevant theories, concepts, and issues. Minimal research and use of established methodologies, leading to incomplete knowledge of the subject. Significant errors and missing components are apparent. Little to no use of learning resources. Arguments are very weak, with no indication of alternative perspectives. Little to no evidence of originality, creativity, or problem-solving skills. Expression and presentation do not meet the necessary standards for accuracy and proficiency. Communication and expression are severely lacking, with significant gaps in professional skills.
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